A COMPREHENSIVE REPORT

ON

Vehicle Ground Clearance for Indian Speed Bumps

By

Name of the student	ID No.
Pushkaraj Ambalgi	2020A4PS1985G

AT

Mercedes-Benz Research & Development India (MBRDI)

A Practice School II Station of

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

(June 2024)

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Pushkaraj Ambalgi	2020A4PS1985G	B.E. Mechanical

Prepared in the partial fulfillment of the
Practice School II Course

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(June 2024)

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
PILANI (RAJASTHAN)

Practice School Division

Station: Mercedes-Benz Research & Development India (MBRDI) **Centre:** Bengaluru

Duration: 5 months **Date of Start:** 10th January 2024

Date of Submission: 19th June 2024

Title of the Project: Vehicle Ground Clearance for Indian Speed Bumps

ID No./Name/: 2020A4PS1985G/ Pushkaraj Vinay Ambalgi
Discipline of the Student: B.E. Mechanical

Name(s) and Designation(s) of the expert(s): Amit Sharma, Uday Udapi, Digvijay Wagh

Name(s) of the PS Faculty: Dr. Shashank Mohan Tiwari

Signature of Student: Pushkaraj Ambalgi Signature of PS Faculty

Date: 19th June 2024

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE

PILANI (RAJASTHAN)

PRACTICE SCHOOL DIVISION

Response Option Sheet

Station: Mercedes-Benz Research & Development India (MBRDI) Centre: Bengaluru

ID No. & Name: Pushkaraj Ambalgi (2020A4PS1985G)

Title of the Project: Vehicle Ground Clearance for Indian Speed Bumps

Usefulness of the project to the on-campus courses of study in various disciplines. Project should be scrutinized keeping in view the following response options. Write Course No. and Course Name against the option under which the project comes.

Refer Bulletin for Course No. and Course Name.

Code No.	Response Option	Course No.(s) & Name
1.	A new course can be designed out of this project.	No
2.	The project can help modification of the course content of some of the existing Courses	No
3.	The project can be used directly in some of the existing Compulsory Discipline Courses (CDC)/ Discipline Courses Other than Compulsory (DCOC)/ Emerging Area (EA), etc. Courses	No
4.	The project can be used in preparatory courses like Analysis and Application Oriented Courses (AAOC)/ Engineering Science (ES)/ Technical Art (TA) and Core Courses.	No
5.	This project cannot come under any of the above-mentioned options as it relates to the professional work of the host organization.	Yes

Signature of Student Signature of Faculty

Date: 19th June 2024 Date:

ACKNOWLEDGEMENT

I would like to express my sincere regards to Mr. Amit Sharma (my reporting manager), for giving me such wonderful opportunity to work at Mercedes-Benz Research & Development India (MBRDI) as an intern. I would also like to thank Mr. Uday Udapi and Mr. Digvijay Wagh (my mentors) for guiding me right from the start. I thank them for extending their valuable guidance, support and for critical reviews of the project and above all, the moral support and motivation they provided me with during all stages of the project.

A special thanks to Dr. Shashank Mohan Tiwari, Faculty-in-charge at my PS II station for guiding me to work at this esteemed organization. Last but not the least, I would like to thank the Practice School Division, BITS Pilani for providing me with such valuable exposure.

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INTRODUCTION

Mercedes-Benz Research & Development India (MBRDI) leverages India's rich pool of engineering and IT talent to drive innovation in both local and global markets. At MBRDI, our focus spans a wide array of areas, including computer-aided design and simulations (CAD and CAE) for powertrain, chassis, and exteriors, as well as embedded systems, telematics, and the development of diverse IT applications and tools.

I am a member of IDD department here at MBRDI and I work with the Chassis and Vehicle Dynamics (CVD) team in the Lateral and Vertical Dynamics vertical. I work with the chassis simulation team where the main objective is to improve the chassis behaviour for special load cases and suggesting countermeasures during pre and series development.

The project that I will be working on is Vehicle Ground Clearance for Indian Speed Bumps The main objective of the project is to obtain maximum dimensional limits for a speed bump of different shapes. This project is based on the concept of multibody dynamics. The advantages that will result from this project are as follows: -

a) Adaptation to local road conditions, especially speed bumps.
b) Improved passenger comfort and safety.
c) Enhanced vehicle durability in diverse road environments.
d) Increased market competitiveness in India.
e) Compliance with local regulatory standards.
f) Positive impact on brand image.
g) Potential for reduced maintenance costs.

So, this project will not only addresses the immediate challenge of adapting to Indian road conditions but also contributes valuable methodologies and insights for future vehicle design and simulation endeavours.

PROJECT TIMELINE

The project timeline unfolds in a structured manner. As you can see in Fig1. The table consists of two columns, the main task and the sub task on the left and next to each task the planned timeline is shown in yellow. These tasks will be marked green once they are completed. This project will begin with a comprehensive understanding of Simpack and the preparation of road files. Subsequently, the focus shifts to model preparation in the following month, encompassing functional data readiness, load case studies, and thorough post-processing of results. Moving into the subsequent month, attention will be directed towards the Exoknox tool, where familiarity with the software interface is established. The tool will be used for generation of the functional data files for bushing, damper, and bump stop. The subsequent week will be dedicated to model simulation and post-processing, culminating in the initiation of ground clearance load case simulations on multiple vehicle carlines in the month that follows. The final week of the project will be reserved for rigorous validation of the results, ensuring accuracy and reliability in our findings. Documentation for the Exoknox tool will occur concurrently with ongoing project tasks. Weekly reviews by my mentors will ensure continuous feedback, and monthly presentations will showcase partial completion milestones.

Sr. no.	Main task	Sub task	Timeline																									
			February		March				April				May					June				July			
			CW8	CW9	CW10	CW11	CW12	CW13	CW14	CW15	CW16	CW17	CW18	CW19	CW20	CW21	CW22	CW23	CW24	CW25	CW26	CW27	CW28	CW29	CW30
1	Introduction to CVD team	Team Presentation																							
2	General Know How	Multi body dynamics																							
		Simpack tutorials and simpack template study																							
		Road file for Simpack																							
	Model Preparation	Functional data preparation(bushing, dampers, bumpstops etc.)																							
		Load cases study																							
		Post-processing of results																							
3	EXONOX	EXONOX tool know how																							
4	Model Simulation & Post-Processing	FKE load case simulation with correlation																							
		Ground clearance load case simulations																							
5	Carlines Simulations	Ground clearance load case simulations - Carline1																							
		Ground clearance load case simulations - Carline2																							
		Ground clearance load case simulations - Carline3																							
6	Final Results and Validation	All simulated carlines post-processing																							
7	Report and presentation	Model Building																							
		EXOKNOX																							
		Ground clearance simulation																							
		Summary for all carlines																							
		Final Report																							

Figure 1:Project Timeline

MULTIBODY DYNAMICS FOR VEHICLE GROUND CLEARANCE

Multibody dynamics is a branch of mechanics that focuses on the study of the motion and interaction of multiple interconnected bodies. By understanding the complexities of interconnected systems in motion, insights can be gained into the behaviour of mechanical systems, optimize designs, and improve performance across a wide range of industries. Multibody dynamics offers a powerful framework for understanding and analysing the motion and interaction of interconnected bodies in mechanical systems. By applying principles of kinematics, dynamics, and numerical methods, engineers can address a wide range of challenges in robotics, automotive engineering, aerospace, biomechanics, and beyond.

Multibody dynamics applications for ground clearance optimization encompass various aspects of vehicle design and performance. The various components that are considered during the multibody analysis are shown in Figure 2. These includes suspension geometry, chassis, tire, drivetrain, vehicle load, and road profile. Fine-tuning suspension geometry, adjusting damping characteristics, and optimizing vehicle packaging to maximize ground clearance while maintaining ride comfort and stability are some of the parameters that are varied during the analysis. Multibody dynamics studies bushings and bump stops by modelling their nonlinear characteristics, including stiffness, damping, and compression. This analysis ensures accurate simulation of how these components influence suspension motion and impact absorption, allowing engineers to optimize ground clearance while maintaining ride comfort and stability.

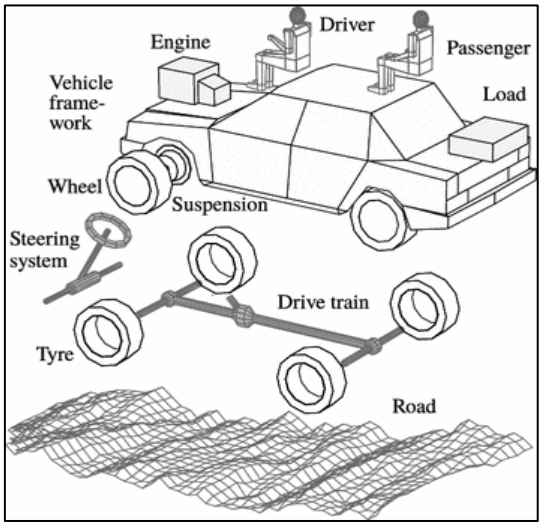


Figure 2:Multibody Dynamics

SOFTWARES/TOOLS

Simpack

SIMPACK is a robust 3D multibody simulation tool designed to analyse the behaviour of mechanical systems. It can accurately simulate the motion, forces, and accelerations of complex multibody systems. With SIMPACK, user can build detailed models, import CAD and FEM data, and conduct analyses in both time and frequency domains. It is particularly adept at modelling elastic bodies and contact mechanics, making it suitable for a wide range of engineering applications requiring precise simulation of mechanical systems.

In Simpack, I will utilize existing templates to construct a model and accompanying test rig. This software enables me to define various parameters of the vehicle, including its mass properties, force elements for different bodies, and the location of hardpoints. Simpack offers three distinct working modes: Simpack-Pre[Figure 3] to build the templates, Wizard is used to parameterize based on actual vehicle data and then to simulate standard loadcases from existing database. Figure 5 shows the workflow of Wizard. The main model and loadcases are built using the existing components and then a scenario is built to carry out the simulation. DOE

(Design of Experiment) is the third working mode of Simpack. Additionally, Simpack's post-processing tool facilitates result visualization, graph plotting, and execution for comprehensive analysis as shown in Figure 4.

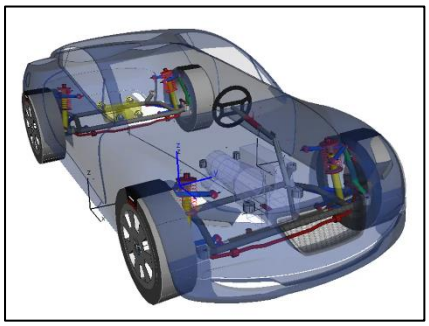


Figure 4: Simpack Pre

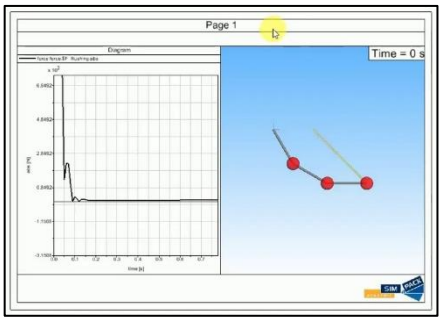


Figure 3: Simpack Post

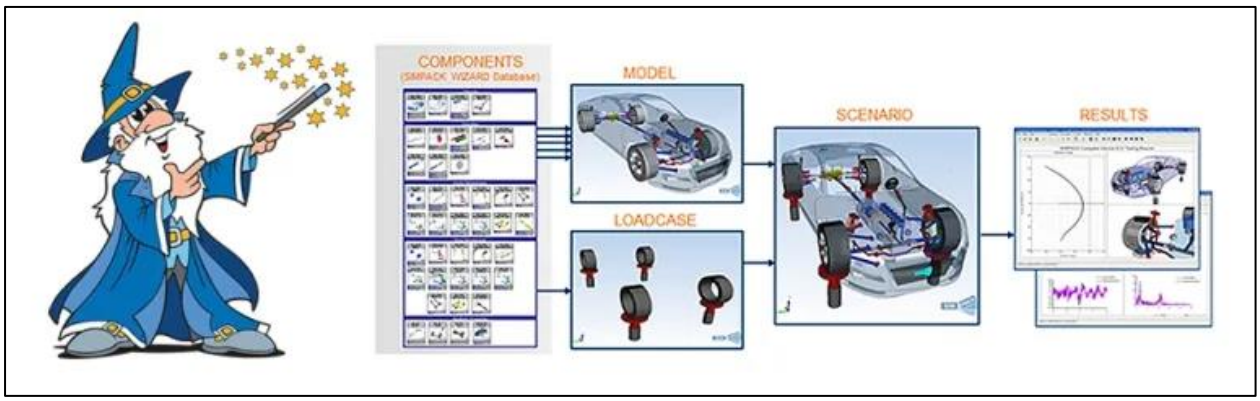


Figure 5: Simpack Wizard

MATLAB

MATLAB is a multi-paradigm programming language and numeric computing environment developed by MathWorks. It enables users to perform matrix manipulations, plot functions and data, implement algorithms, create user interfaces, and interact with programs written in other languages.

MATLAB will be used for two purposes in the project. Firstly, it will be used for the .crg file creation through OpenCRG source. OpenCRG is a comprehensive project available for free and open-source use, focusing on the creation, modification, and assessment of road surfaces. It introduces the CRG (curved regular grid) file format specification, aiming to standardize detailed descriptions of road surfaces. This format finds applications in tire, vibration, and driving simulations. Additionally, it offers optional features such as hilliness (slope, inclination, grade, and pitch angle) and cross slope (super-elevation, banking, cant, and roll angle). A sample of road profile is shown in Figure 6. The primary objectives of the CRG file format specification are compact storage and adequate accuracy, even for lengthy tracks. The coding of the road file will be done in MATLAB.

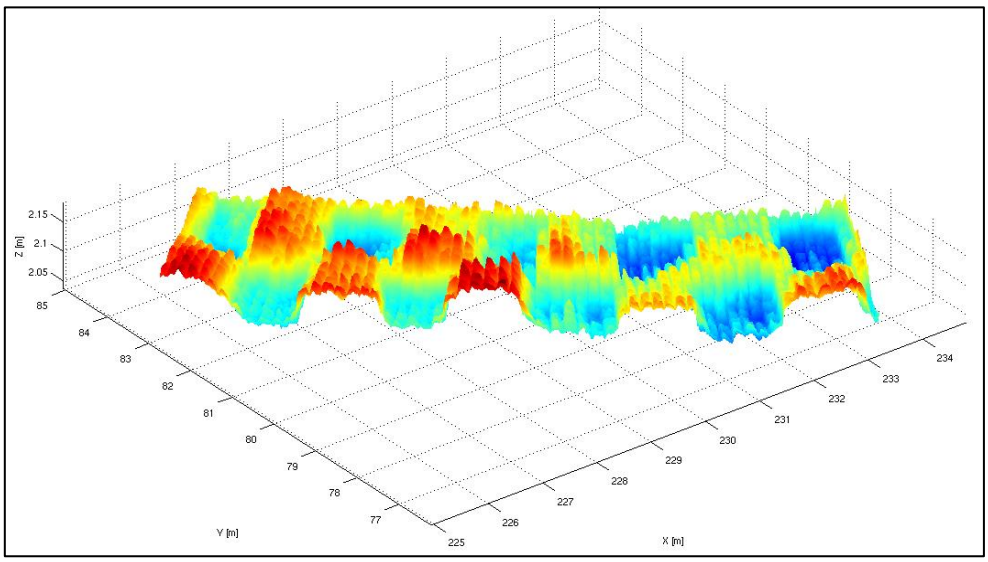


Figure 6:CRG File

The secondary function of MATLAB within this project entails executing simulations through script execution. This process involves invoking the main model file and subsequently adjusting sub-variables such as mass properties, force elements, and load case conditions via script. Upon finalizing all requisite properties, the load case file is invoked, followed by the execution of the designated load case.

Exoknox

EXOKNOX was developed by automotive industry calculation engineers in collaboration with experienced software engineers from Karakun. The main aim of Exoknox is to

- Break down silos with the objective of contextualizing the stored information and knowledge.
- Enable engineers to focus on actual development work - saving time on data collection.
- Shift the focus towards product maturity development.

The functional data of the components like bushing, bumpstop, damper etc. from the suppliers is present in the raw form. The data is then fed to Exoknox to get the data organized in the format as required by Simpack. Refer Figure 7 for the description. Firstly, a part number is searched in Exoknox at the top left corner. This will provide the main bundle of the file in the bottom tab. The main bundle consists of the target data, order and delivery with supplier, and experimental testing data from supplier. Graphical representation of the functional data is also available in Exoknox as can be seen in the central tab of the figure 7.

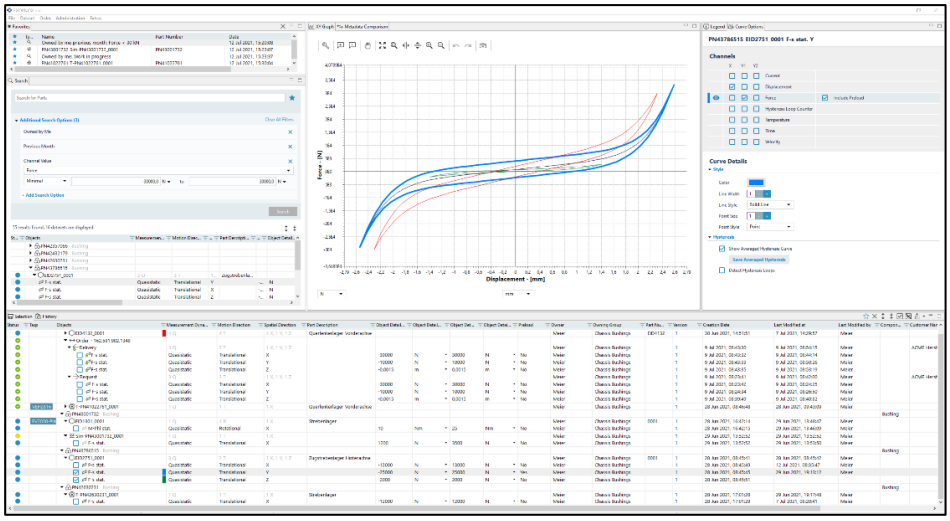


Figure 7:Exoknox

Tool (Confidential)

This tool is a proprietary tool based on MATLAB developed by engineers in Mercedes-Benz, Germany for internal use. The tool is highly confidential and not open for public.

PROJECT SETUP

Boundary Conditions

According to ARAI Automotive Research Association of India there are three different types of loading conditions of a passenger vehicle. These loading conditions are made to check the behavior of vehicle under different conditions. The three different types are as follows.

1) Basic loading condition – It consists of the curb weight of vehicle along with one passenger (75 kgs each).
2) Design loading conditions – It consists of the curb weight of vehicle with 50% of fuel along with three passengers (75 kgs each).
3) Full loading condition – It consists of curb weight with 100% fuel capacity with all passengers along with the boot space and accessories.

Figure 8 provides a simplified equations for boundary conditions.

1) Basic load – Curb Weight + No Fuel + 1 passenger (75kgs each).
2) Design load – Curb Weight + 50% Fuel + 3 passengers(75kgs each).
3) Full load – Curb Weight + 100% Fuel + all passengers + boot space and accessories.

Figure 8: Boundary Conditions

Basic Loadcases

Driving Straight

In the driving straight loadcase the vehicle is given a run with a constant velocity over a straight road as shown in figure 9.

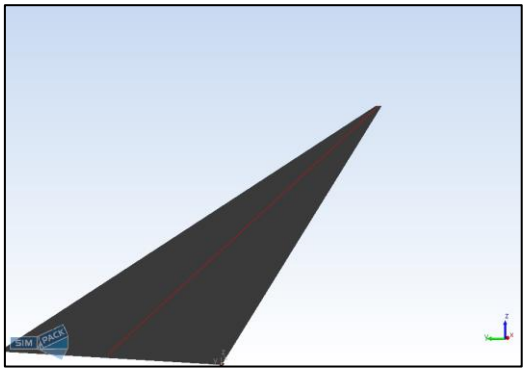


Figure 9: Driving Straight loadcase

As shown in figure 10 the longitudinal and vertical dynamics of the vehicle are studied. In the longitudinal dynamics the velocity and longitudinal acceleration of vehicle are plotted against the time. This is important to know how much time the vehicle takes to reach its steady state. Secondly the vertical dynamics consists of plot of wheel travel and tire force against time. Wheel travel signifies how much does the vehicle dive or squat. This can be also used to tell whether the vehicle has reached a steady state or not.

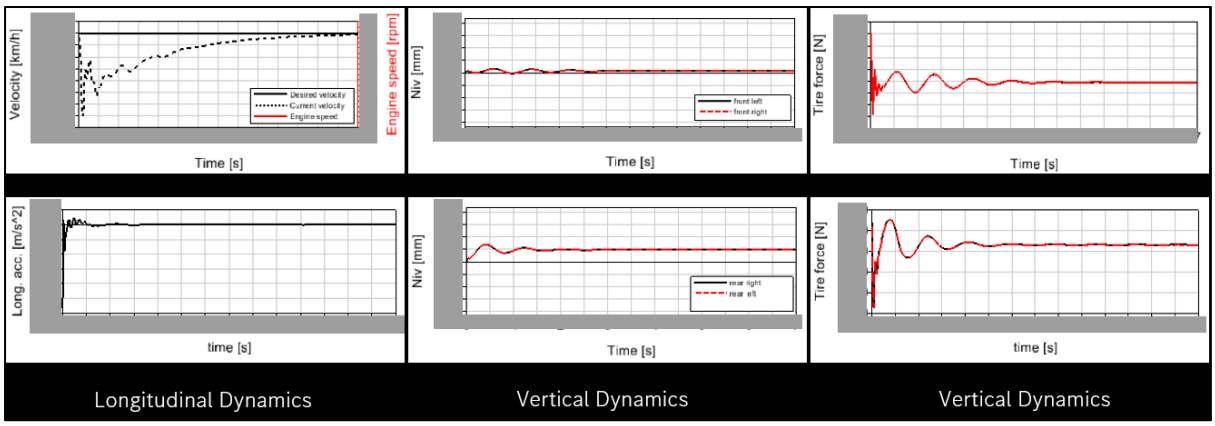


Figure 10: Driving Straight Graphs

Kinematics and Compliance

In Kinematics and Compliance loadcase the vehicle is placed on 4 pedestal such that each of its wheel is on a hydraulic piston. Then the pistons are moved up and down either in parallel or opposite sense for the rear and front side and the motion of wheel is observed. The test rig is shown in figure 11.

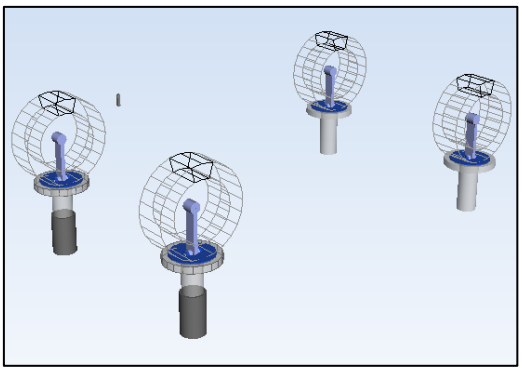


Figure 11: Kinematics and Compliance test rig

Figure 12 shows four important graphs that are studied for this loadcase. This include spring stiffness v/s the wheel travel. Secondly the wheel load is plotted against the wheel lift. Thirdly the camber angle (it is the angle between the vertical axis of a vehicle and the vertical axis of the vehicle when viewed from the front or rear) is plotted against vertical wheel position in vertical direction. And fourthly the toe angle (is the symmetric angle that each wheel makes with the longitudinal axis of the vehicle) is plotted against the wheel position in vertical direction.

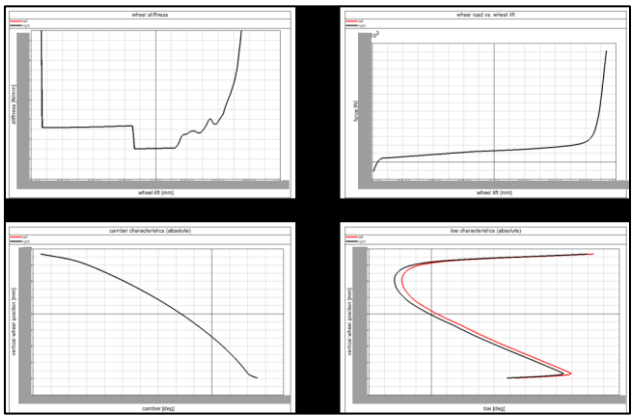


Figure 12: Kinematics and Compliance

Ground Clearance

In the Ground Clearance loadcase the vehicle is made to pass over different road profiles which includes a crest, a dip, and a curbstone(step) as shown in figure 13. Here the main aim is to check whether the car underbody hits the road surface or not.

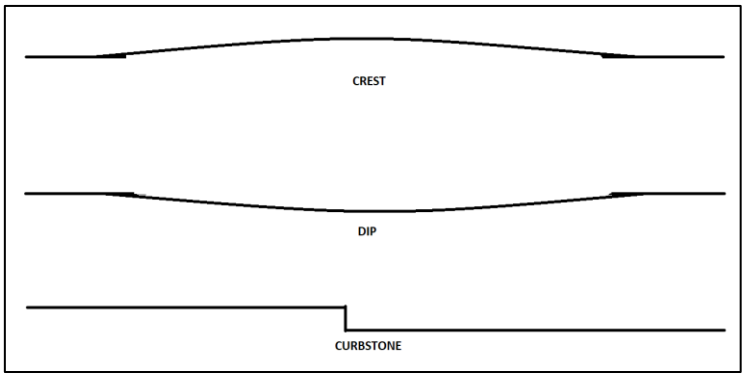


Figure 13: Road profiles for Ground Clearance.

The graphs similar to driving straight loadcase are studied for the ground clearance loadcase as shown in figure 14.

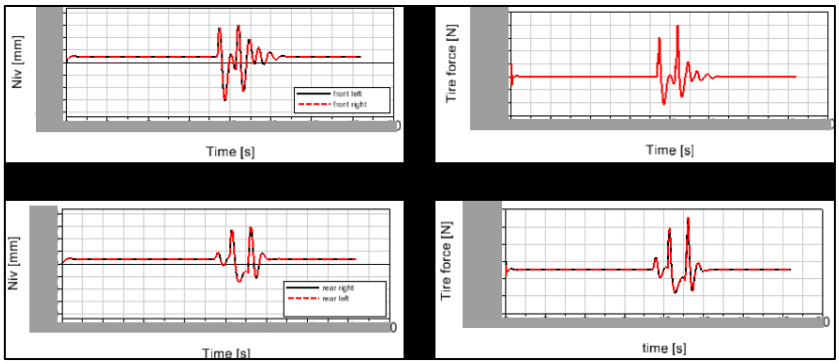


Figure 14: Ground Clearance

One important plot that the ground clearance loadcase postprocessing generates is the grenzeflaeche plot. It is the graph in which y axis is the vertical direction of the vehicle and x axis is the longitudinal direction of the vehicle. The plot gives a contour of points which denotes the minimum distance between vehicle and road profile for all longitudinal points on the vehicle that it may encounter during a complete run. It helps to show how close road surface can get to a car underbody. A sample grenzeflaeche plot is shown in figure 15.

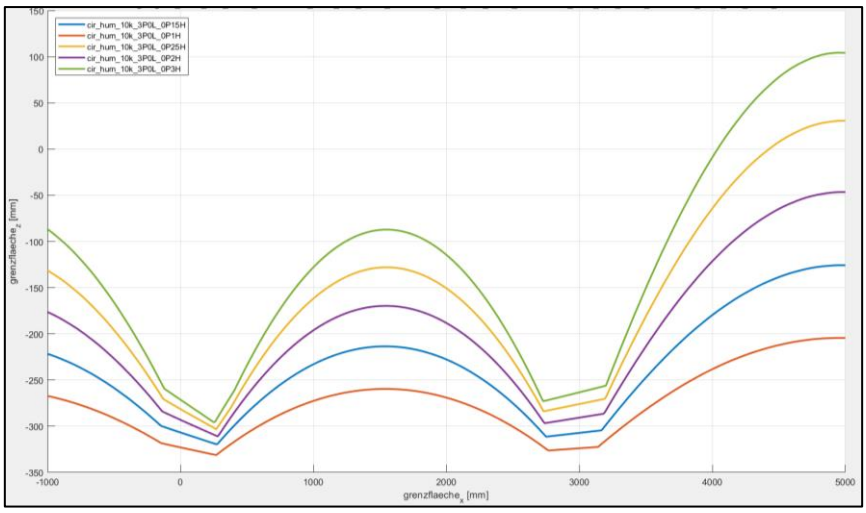


Figure 15: Grenzeflaeche plot

MATLAB scripting

This project entails executing simulations through script execution in MATLAB. Firstly, the location of the work folder is set, followed by the main model being called. The model is then reset to default, and the scenario and loadcase variant are named. Subsequently, the required loading condition, whether it be Basic, Design, or Full load, is applied. The necessary subvars are adjusted accordingly. Next, the adjusted subvars are set to the model. Finally, the MATLAB script is then executed.

Car underbody

The underbody refers to the belly of your car, the underside that sits closest to the road. Everyday as we drive, rocks, road debris, and salt on snowy roads brushes up against the underbody of your car. Thus a car underbody is a important factor while analysing the ground clearance of a vehicle. In a car underbody plot the x axis is the longitudinal direction of vehicle and y axis is the vertical direction of the vehicle. The plot consists of the lowermost points of the vehicle for its entire longitudinal length. Figure 16 shows a sample plot of a car underbody. The vehicle is divided in 3 section – front, rear, and middle and the various parts of the car under body are shown in figure 16.

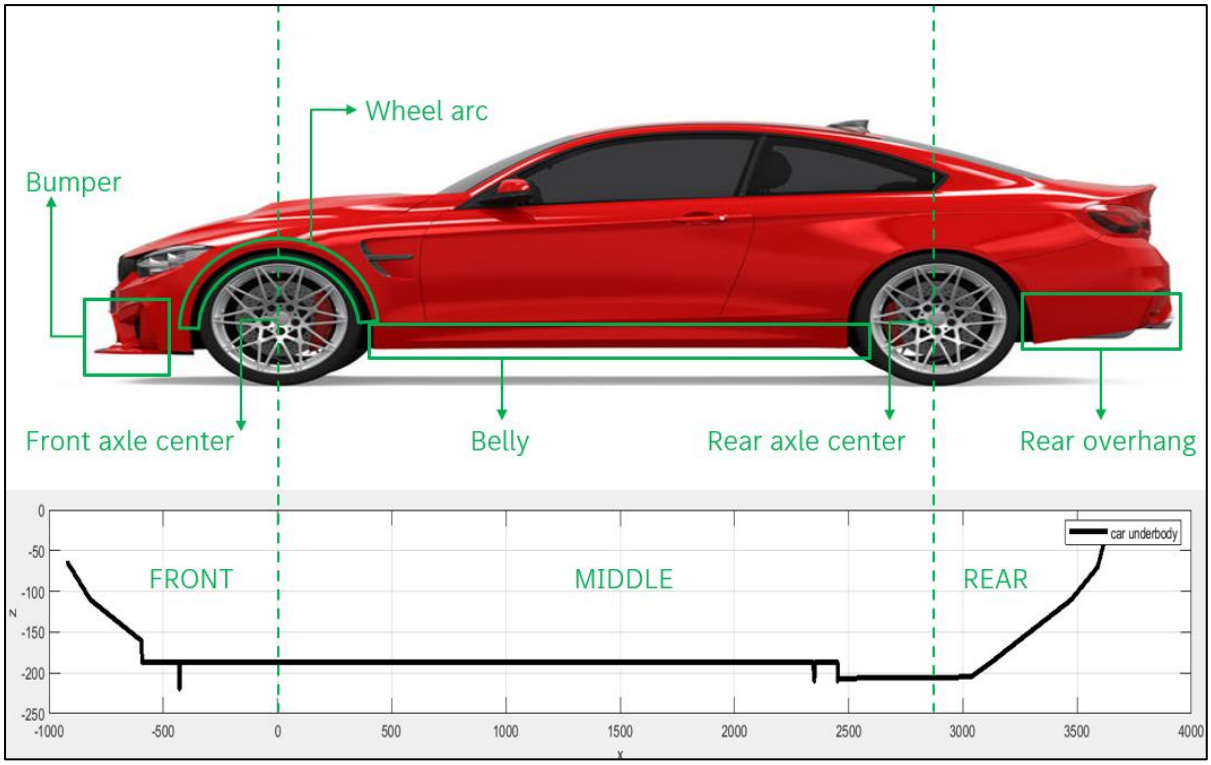


Figure 16: Car underbody

CRG File creation

```
1)  run("T:\TestDrive\CRG\OpenCRG_V1.1.2_release_by_VIRES_Simulationstechnologie_GmbH\OpenCRG.1.1.2\matlab\crg_init.m");
    dispRes = 0;
2)  uinc = 0.005;
    vinc = 0.005;
    nv = 711;
    nu = 53761;
    v = -(nv-1)/2*vinc:vinc:(nv-1)/2*vinc;
3)  z= zeros(nu,nv);
    data.u = (nu-1)*uinc;
    data.v = (nv-1)*vinc/2;
    data.z = z;
4)  d = 0.225;
    h = 0.08;
    R = (d^2+h^2)/(2*h);
    L_Welle80mm = .450;
    theta1 = asind(1-h/R);
    theta2 = 180-theta1;
5)  laenge_Welle80mm = 2*R*sind(90-theta1);
    if (laenge_Welle80mm - L_Welle80mm) > 0.1
        error('Given length of bump does not match with calculated length');
    end
6)  steps = 0:uinc:L_Welle80mm; % (nu-1)/10
    groesse_steps = size(steps,2);
    theta = linspace(theta1,theta2,groesse_steps);
    Z_bump = h-R*(1-sind(theta));
    figure
    plot(steps,Z_bump)
    axis([0 .450 0 0.45])
7)  m_start_1 = 70; % m
    m_end_1 = m_start_1 + 0.45; % m
    u_start_1 = (m_start_1)/(uinc*nu/nu)
    u_end_1 = u_start_1 + (m_end_1-m_start_1)/(uinc*nu/nu)
    for j =1:nv
    data.z(u_start_1:u_end_1,j) = Z_bump';
    end
8)  figure
    plot(uinc:uinc:nu*uinc,data.z(1:size(data.z,1),1),'LineWidth',3,'Color','#FF8800')
    legend('Strasse mit SP80mm')
    xlabel('X[m]')
    ylabel('Z[m]')
    grid on;
    axis([-0.02 nu*uinc -0.05 0.5*nu*uinc]);
9)  data.ct{1} = 'CRG defined by z matrix: flat-track-zero elevation with bump of h80mm @ 70m';
    crg_write(crg_single(data), 'sleeping_policeman_demo.crg');
    datRead = crg_read('sleeping_policeman_demo.crg');
    crg_show(datRead);
```

Figure 17: CRG File

Refer figure 17 for the below given description of CRG file creation.

1) Run the folder location of OpenCRG.

2) Then create road struct with uniform increments and then define the length in x and y direction.
3) Create an array to assign the road elevation at every point in x-y plane of the road.
4) Formulate the profile of the speed bump.
5) Validate whether the assigned length of speed bump matches with the calculated length.
6) Plot the speed bump profile.
7) Locate the speed bump on the flat road.
8) Plot the road profile.
9) Create the CRG file.

Standard Speed Bumps given by IRC are generated by using the crg script as shown in figure 18.

Table 3.1 Recommended Radii and Chord Lengths, Circular Humps (Assumed Rise = 10 cm)

Desired Speed	Radius	Chord Length	Bus Speed During Passage
20 km/h	11 m	3.0 m	5 km/h
25 km/h	15 m	3.5 m	10 km/h
30 km/h	20 m	4.0 m	15 km/h
35 km/h	31 m	5.0 m	20 km/h
40 km/h	53 m	6.5 m	25 km/h
45 km/h	80 m	8.0 m	30 km/h
50 km/h	113 m	9.5 m	35 km/h

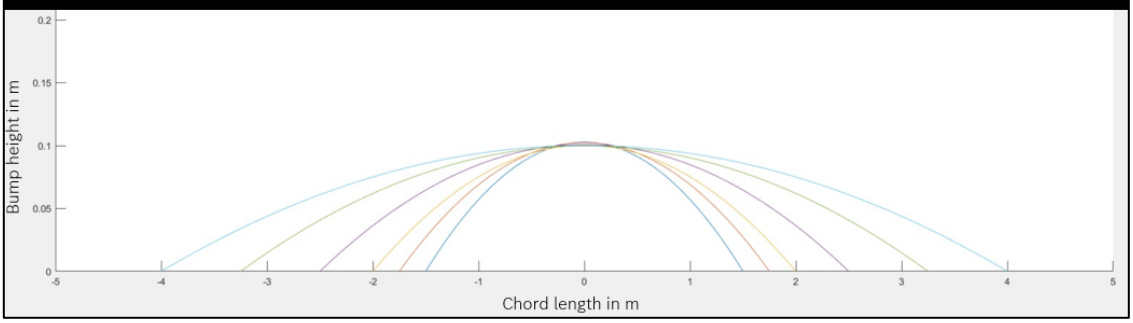


Figure 18: IRC Speed Bumps

Project Methodology

IRC Speed Bumps Simulation

Indian Roads Congress is the premier technical body of highway engineers which was formed in 1934 as India's national body for laying down and designing standards for roads and highway construction and provides a stage for exchanging expertise and latest research developments relating to it. For the ground clearance simulations the standard speed bumps given by IRC were taken first to run the ground clearance loadcase. The standard circular speed bumps given by IRC are shown in figure 19.

Table 3.1 Recommended Radii and Chord Lengths, Circular Humps (Assumed Rise = 10 cm)

Desired Speed	Radius	Chord Length	Bus Speed During Passage
20 km/h	11 m	3.0 m	5 km/h
25 km/h	15 m	3.5 m	10 km/h
30 km/h	20 m	4.0 m	15 km/h
35 km/h	31 m	5.0 m	20 km/h
40 km/h	53 m	6.5 m	25 km/h
45 km/h	80 m	8.0 m	30 km/h
50 km/h	113 m	9.5 m	35 km/h

Figure 19: IRC standard circular bumps

After running the simulations, the results were obtained as shown in figure 20. It was clearly observed that the car maneuvered all the speed bumps easily without the underbody hitting to the road surface as the car underbody plot (black) did not intersect any of the road profiles.

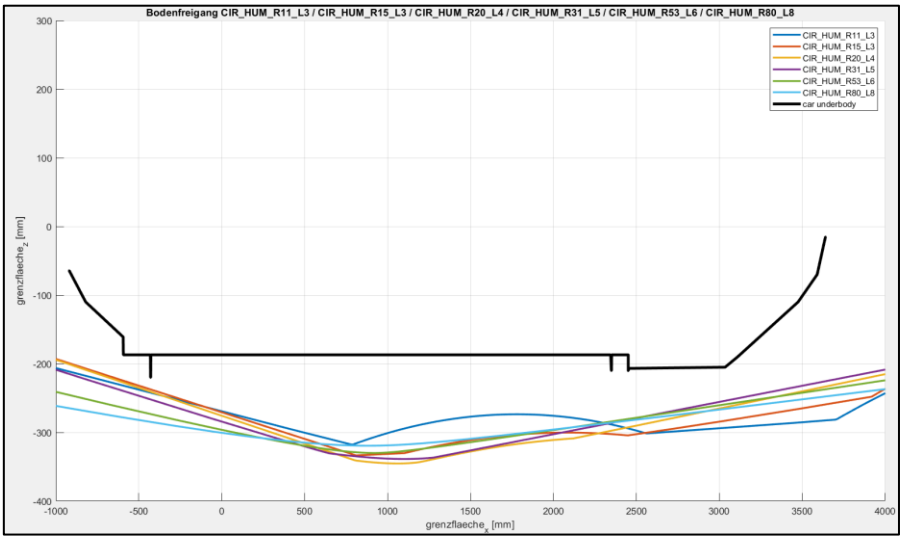


Figure 20: IRC speed bump simulation results with car underbody

Speed Bumps Iterations

Based on the results obtained from the simulations of IRC standard speed bumps it was then decided to run the simulations on much critical speed bumps in which there were chances of the vehicle underbody hitting the road surface. Figure 21 shows the different sizes of speed bumps considered for the ground clearance simulations. The chord length was varied from 0.5m to 3.0m with an increment of 0.5 m. And for each of these chord lengths the height was varied from 0.1m to 0.3m with increment of 0.05m. Figure 22 shows samples of speed bumps with different height and chord length along with a sample wheel size.

Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	0.5	0.1	0.3625
10	0.5	0.15	0.2833
10	0.5	0.2	0.2562
10	0.5	0.25	0.25
10	0.5	0.3	0.2542
Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	1	0.1	1.3
10	1	0.15	0.9083
10	1	0.2	0.725
10	1	0.25	0.625
10	1	0.3	0.5667
Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	1.5	0.1	2.8625
10	1.5	0.15	1.95
10	1.5	0.2	1.5063
10	1.5	0.25	1.25
10	1.5	0.3	1.0875
Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	2	0.1	5.05
10	2	0.15	3.4083
10	2	0.2	2.6
10	2	0.25	2.125
10	2	0.3	1.8167
Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	2.5	0.1	7.8625
10	2.5	0.15	5.2833
10	2.5	0.2	4.0062
10	2.5	0.25	3.25
10	2.5	0.3	2.7542
Speed (km/h)	Length (m)	Height (m)	Radius (m)
10	3	0.1	11.3
10	3	0.15	7.575
10	3	0.2	5.725
10	3	0.25	4.625
10	3	0.3	3.9

Figure 21: Speed Bump Iterations

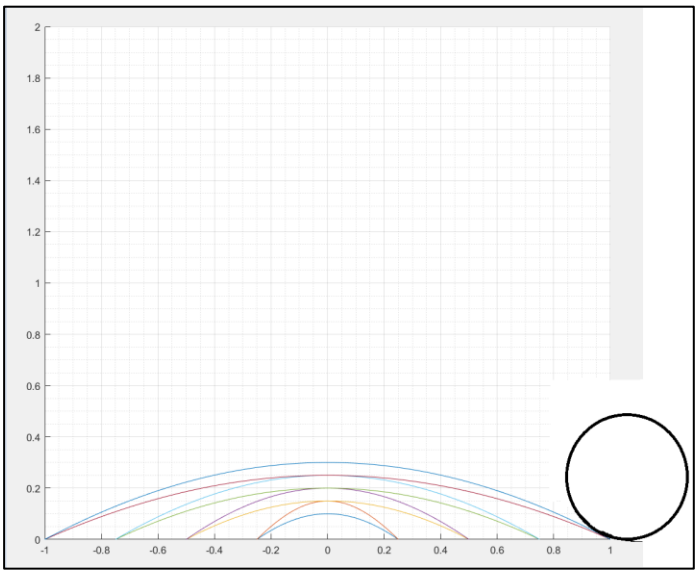


Figure 22: Sample speed bumps

Speed Bump Simulations

The simulations were executed for different sizes of speed bumps. The corresponding grenzeflache plots for different speed bumps were obtained and then they were plotted along with the car underbody as shown in figure 23. The black line represents the car underbody.

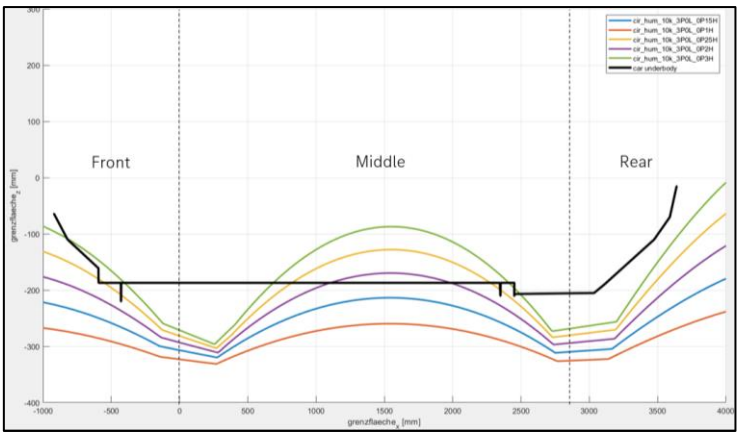


Figure 23: Grenzeflaeche plot with car underbody

The points where the black line intersects the different curves for speed bumps is the point where the car underbody hits the road surface. Based on these plots it was observed that the speed bumps that were considered earlier had a greater height increments and hence it was not possible to conclude at what dimensions of speed bump did the car underbody touched the road surface. Therefore a new set of speed bumps were designed where the height increment is much finer (0.01m) to get the exact shape of speed bump where the car underbody just touches the road surface. The simulation results using these speed bumps are shown in figure 24. And this was done for the front, rear and middle portion of the vehicle. This would help to define the design space to the designer about the maximum limit of speed bump that the front, middle, and rear portion individually can maneuver without hitting the road surface.

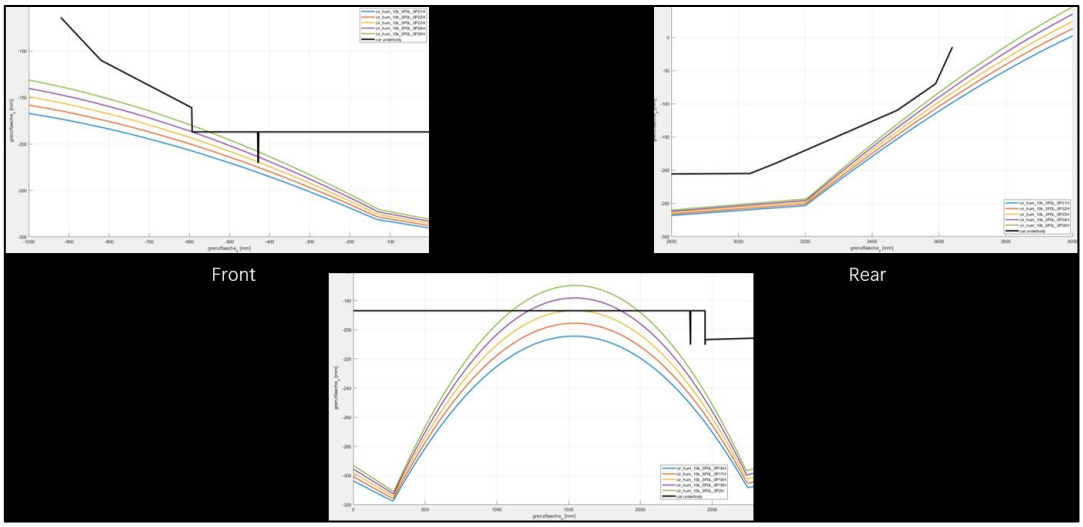
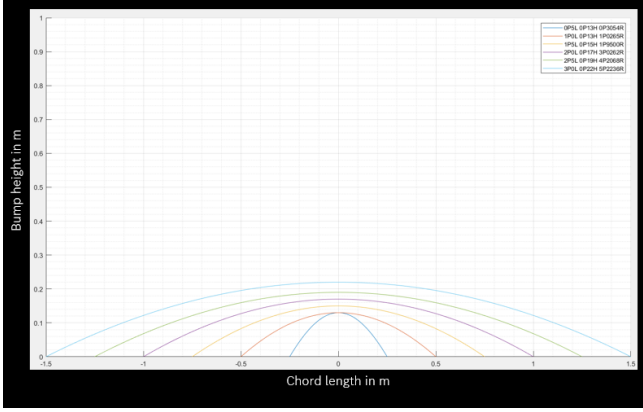


Figure 24: Fine-tuned simulation results

Results

The final outcome of the project was to find the maximum dimensional limits of the speed bumps for Indian road conditions. Based on the simulation results the dimensional limits of the speed bump that a car can maneuver without the car underbody hitting the road surface were obtained for 3 different carlines for 3 different speeds i.e. 10kmph, 15kmph, and 20kmph and for front, middle, rear, and the complete underbody of the car. A sample of results is shown in figure 25, 26, 27, and 28 for front, middle, rear, and car underbody as a whole.

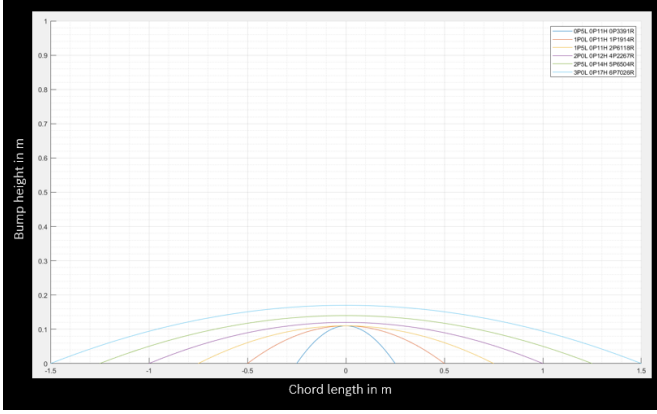
1) FRONT



Speed (in kmph)	Chord Length (in m)	Max. Height (in m)	Radius (in m)
10	0.5	0.13	0.3054
10	1	0.13	1.0265
10	1.5	0.15	1.95
10	2	0.17	3.0262
10	2.5	0.19	4.2068
10	3	0.22	5.2236

Figure 25: Speed Bump Dimensional Limits for Front portion

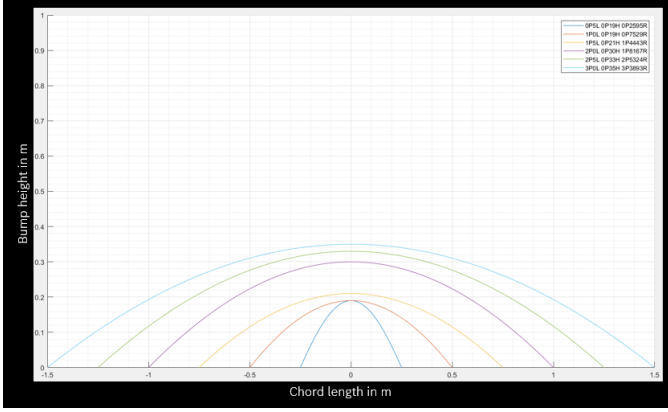
2) MIDDLE



Speed (in kmph)	Chord Length (in m)	Max. Height (in m)	Radius (in m)
10	0.5	0.11	0.3391
10	1	0.11	1.1914
10	1.5	0.11	2.6118
10	2	0.12	4.2267
10	2.5	0.14	5.6504
10	3	0.17	6.7026

Figure 26: Speed Bump Dimensional Limits for middle portion

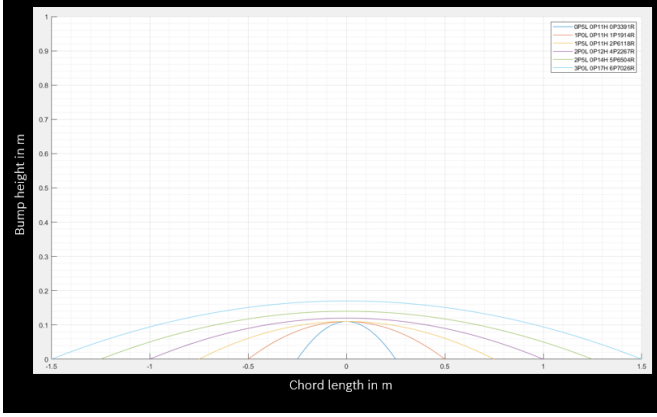
3) REAR



Speed (in kmph)	Chord Length (in m)	Max. Height (in m)	Radius (in m)
10	0.5	0.19	0.2595
10	1	0.19	0.7529
10	1.5	0.22	1.4443
10	2	0.3	1.8167
10	2.5	0.33	2.5324
10	3	0.35	3.3893

Figure 27: Speed Bump Dimensional Limits for rear portion

4) CAR UNDERBODY AS A WHOLE



Speed (in kmph)	Chord Length (in m)	Max. Height (in m)	Radius (in m)
10	0.5	0.11	0.3391
10	1	0.11	1.1914
10	1.5	0.11	2.6118
10	2	0.12	4.2267
10	2.5	0.14	5.6504
10	3	0.17	6.7026

Figure 28: Speed Bump Dimensional Limits for car underbody as a whole

Finally, the results obtained are provided to the design team to develop an idea for the design space considering the ground clearance over the speed bumps in Indian scenario.

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- Figure 6 – Hhelmich. (2010). *OpenCRG belgianblock.* https://en.wikipedia.org/wiki/OpenCRG#/media/File:OpenCRG_belgianblock.png
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- Figure 8 – Self
- Figure 9 – Simpack
- Figure 10 – Simpack Post
- Figure 11 – Simpack
- Figure 12 – Simpack Post

- Figure 13 – Self
- Figure 14 – Simpack Post
- Figure 15 – MATLAB
- Figure 16 – MATLAB
- Figure 17 - Ahmet Rodoplu (2019). *ECE R48, Defined loading conditions for M1 category vehicles*

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- Figure 18 – Indian Road Congress (2018). *Recommended Radii and Chord Lengths, Circular Humps (Assumed Rise = 10 cm)*

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- Figure 19 – MATLAB
- Figure 20 – Self
- Figure 21 – MATLAB
- Figure 22 -MATLAB
- Figure 23 – MATLAB
- Figure 24 - MATLAB